

FUNNY STORY: THE EFFECTS OF
BACKGROUND LAUGHTER ON HUMOR
PROCESSING IN THE BRAIN

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Abstract

Does hearing other people laugh actually make something funnier? This paper attempts to better understand how our brains respond to auditory stimuli that contain background laughter as opposed to those that don't. I compare fMRI data from participants listening to the same story told with and without audience laughter in the background. Using a General Linear Model (GLM), I find significantly greater activation in the bilateral superior temporal gyri (lSTG, rSTG), Heschl's gyri, and the pre-supplementary motor area (pre-SMA) in the group that heard background laughter, even under stringent family-wise error rate (FWER) correction via non-parametric permutation testing. These regions have established roles in humor comprehension, appreciation, and the physical expression of laughter. A complementary Multi-Variate Pattern Analysis (MVPA) confirms that punchline events can be decoded above chance across all regions of interest in both groups, though no statistically significant between-group differences in classification accuracy are found. Together, these results suggest that the mere sound of laughter in the background of humorous story amplifies neural responses associated with humor comprehension and appreciation, offering a neuroscientific account of the contagiousness of laughter.

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For the wonderful people who bring humor into my life.

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Chapter 1

Introduction

Have you ever been annoyed watching a show or listening to something with an obnoxiously loud laugh track? Or, on the contrary, have you noticed yourself genuinely laughing along to something you wouldn't have otherwise found as funny just because other people around you were laughing? In either case, you wouldn't be alone. Classic shows like *Friends*¹ and *Seinfeld*² have been criticized online for over-relying on laugh tracks to evoke an emotional response without actually providing much comedic content. You may have come across a clip³ of a popular show with the laugh track removed, highlighting all the awkward pauses and lackluster punchlines. The contagiousness of laughter is a well known phenomenon, with research long ago confirming that just the sound of laughter itself evokes laughter[1]. These behavioral effects of hearing laughter lead to questions of how responses to it are reflected in brain activity. Which parts of our brains light up when we listen to the same story with and without laughter in the background? Does hearing laughter have an impact on our cognitive state beyond just registering the auditory signal? The primary aim of this thesis is to explore that possibility.

¹https://www.reddit.com/r/friends_tv_show/comments/177tjo7/friends_is_not_funny_without_the_laugh_track/

²<https://decider.com/2015/07/22/seinfeld-laugh-track/>

³<https://www.youtube.com/watch?v=8ib889nSZVU>

In neuroscience, functional magnetic resonance imaging (fMRI) imaging is a highly popular, powerful, and non-invasive tool that has been used as a way to measure brain activity since the early 1990s [2]. In the decades since, the technology has come incredibly far, with new methods and exciting results being published every year. fMRI data is collected from individuals in a scanner, and reflects a measure of activation called the BOLD signal for thousands of voxels (volumetric pixels) in the brain [2]. In most experiments, this signal is sampled frequently to capture changes in activity over time.

Using established fMRI analysis methods, I will be looking at fMRI data collected from individuals listening to engaging, comically rich short stories. Some of them were recorded in front of a live studio audience with audible background laughter, while others were recorded in a silent studio. The fact that individuals were only listening to recordings of these stories allows for the isolation of effect of the sound of laughter, as opposed to fMRI data collected from TV show or movie viewing.

Most notably, I will analyze scans from participants who listened to the same story told by the same speaker under the two different conditions. Two different sets of fMRI study participants listened to almost identical versions of Jim O’Grady’s *Pie Man*: one told at a live storytelling event, one recorded while he himself was undergoing scanning[3]. These two studies were done at different times at the Princeton Neuroscience Institute and later published as a part of an aggregated narrative storytelling dataset[3]. This presented a unique opportunity for a direct comparison of the data from both, and was the initial inspiration for this project.

Chapter 2

Background and Related Work

2.1 Humor

Humor is a critical, multifaceted tool for human communication[4]. We use it to bond[5], for tension relief, and even as a device for political dissent[6]. Humor has been studied in all its forms across the social sciences[4, 5, 7], humanities [8], and even biological sciences[9].

Humor widely permeates social dynamics, but we are also able to experience mirth by ourselves. In the age of online streaming especially, funny content is abundant and easily accessible for the lone consumer. We put on a sitcom in the background while folding laundry, we listen to comedy podcasts at the gym, and we watch short form stand-up comedy clips during a solo lunch. It's no new observation that this experience differs from engaging with humor in a social setting. Mitschke et al.[10] analyzed laughter regulation in social versus solitary settings. They found that while people are able to suppress amused facial expressions using emotional regulation strategies when by themselves, the laughter of other people makes it harder to do so. There is less we know about how individuals experience and appreciate humor when they are alone.

2.2 fMRI Analysis

While I touched on it briefly in the introduction, more background on how fMRI is used in neuroscience is necessary for understanding the methods in this work.

When your neurons are activated, they expend energy. In order to restore that energy, an increased flow of oxygenated blood is sent to the area. Magnetic resonance imaging machines are then able to pick up on that oxygenation level to obtain a BOLD (blood oxygen level dependent) signal through some fancy physics[11]. It suffices to know that MRI scanners can measure the contrast in concentration of oxygenated versus de-oxygenated hemoglobin (a protein containing iron found in red blood cells) in the brain [12]. After a neuron or clusters of neurons has used its energy, that area receives an influx of oxygenated blood. The BOLD contrast peaks about 5-6 seconds after the expenditure of energy, likely in response to some stimulus. Thus a high BOLD signal detected in a region indicates that it was active a few seconds prior[12]. This delay, termed hemodynamic lag, is the time it takes for the blood to actually get to that part of brain [12]. It is considered one of the main drawbacks of fMRI data as representation of brain activity, and must be accounted for in any fMRI analysis¹.

The BOLD signal is measured at the granularity of voxels, or volumetric pixels. Modern scanners have high voxel resolution, with voxel edge lengths varying from 1.5mm-3mm. For reference, a cubic millimeter contains around 32000 neurons [13]. In most neuroscientific experimental setups involving the collection of fMRI data, the BOLD signal is sampled at different time points. For studies that involve participants looking at blocks of static stimuli, this sampling may be done as the stimulus being presented is switched out. For studies that show participants dynamic stimuli, BOLD is measured at a fixed rate. An interval at which the signal is measured is called a time repetition (TR), and is usually somewhere around a couple of seconds[12]. With

¹Thanks to Prof. Ken Norman for an excellent expository lecture on this in NEU480.

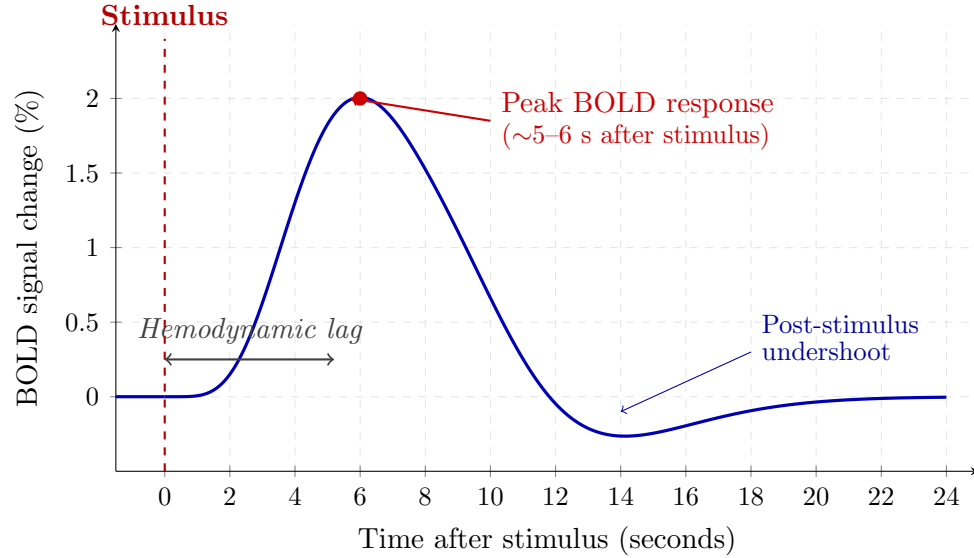


Figure 2.1: The hemodynamic response function (HRF) after exposure to a stimulus at $t = 0$. The BOLD signal peaks approximately 5–6 seconds after neural activity, a delay known as the *hemodynamic lag*, before returning to baseline with a post-stimulus dip.

all these components, fMRI data paints a per-voxel picture of brain activity over the course of time and exposure to stimuli. This data can then be used to learn about activation patterns through a plethora of varied experimental designs and methods.

2.3 Humor Appreciation in Neuroscience

While fMRI data is abundant, relatively little of it has thus far been used in understanding how humor works in the brain. Searches on Neurosynth², a platform that aggregates and meta-analyzes fMRI data from thousands of studies, yield no results for terms such as “humor”, “amusement”, “comedy”, “funny” and “mirth”. Despite this, there has been sufficient groundwork laid in the field to enable the analyses in this study.

Chan et al. [14] and Kennison et al. [15] have reviewed the neuroscience of humor in recent years. In their review, Chan et al. present a theory of humor processing which states that it has three subsequent components: comprehension

²<https://neurosynth.org/>

(the cognition), appreciation (the affect), and expression (the physical response) [14]. This is supported by prior work. In an early study that used cartoons as stimuli, Bartolo et al [9] found that both brain hemispheres, and particularly parts of the frontal³ and temporal⁴ gyri, activated when participants were shown something funny. Further studies have also shown that laughter and mirth are cognitively different, supporting the tri-component theory [14]. In my experiments, I will be utilizing this tri-component framework to see whether background laughter has an significant impact on brain activity beyond difference in expression.

2.4 Regions of interest

For many fMRI analysis techniques, performing them on all the voxels of the brains of dozens of participants is computationally intensive and often unnecessary. Instead, it is conventional in studies to focus on certain regions of interest, or ROIs [9, 16, 17, 18, 19, 20, 21]. Thus, it is important for this work to also define the scope of ROIs. Using Chan’s et al. tri-component framework [14], I have selected ROIs that have been identified as relevant for humor comprehension, appreciation, and expression.

When it comes to laughter (humor expression), a tractography-based fMRI study by Caruana et al. [16] identified two distinct, largely disconnected networks involved in its processing. They concluded that the first network, consisting of the pregenual anterior cingulate cortex (pACC), ventral striatum/nucleus accumbens (VS/NAcc), and ventral temporal pole (TPv), is primarily involved in emotional, involuntary laughter. The second network they found ran between the frontal operculum (FO) and primary motor cortex (M1), which they believed to be tied to voluntary and conversational laughter. In both of these networks, they identified a connection to the pre-supplementary motor area. This suggested that this area is involved in the

³Middle frontal gyrus visualization, BrainFacts.org

⁴Temporal lobe visualization, BrainFacts.org

laughter response in both circumstances and not specific to a reason or emotion. The SMA was also shown to be highly active in an fMRI study of tickling which measured participants brain activity while being tickled and allowed to laugh, being tickled and asked to inhibit laughter, and being asked to laugh voluntarily. Wattendorf et al. found especially high activation in the tickling-and-laughter and voluntary laughter conditions [17]. This work suggests the SMA’s function is tied to the physical response of laughter rather than mirth, and as such I chose it as the key ROI representing humor expression in the brain.

There seem to be more areas involved in humor comprehension and appreciation. Bartolo et al.’s cartoon study looked for areas involved in both of these particular functions [9]. It’s also important to note that participants in this study were asked not to laugh, indicating that these areas, if involved with mirth, are not solely involved with the physical response of laughter. In their study, Bartolo et al. presented participants with pre-labeled not-funny and funny stimuli, asking them to rate each stimulus on a scale from 0 to 6. They identified areas to be connected to humor comprehension by comparing activation while viewing funny and non-funny stimuli (based on prior labeling). These were the right inferior frontal gyrus (rIFG), the left superior temporal gyrus (lSTG), the left middle temporal gyrus (lMTG), and the cerebellum [9]. They also looked at degree of amusement by correlation of activity with participants’ funniness rankings, finding an association with the left amygdala and cerebellum in addition to rIFG, lSTG, and lMTG [9]. The bilateral STG, along with the left Heschl’s gyrus, also showed significantly greater activation when participants listened to genuine laughter in contrast with when they listened to posed laughter in another fMRI study [19].

The pACC, VS, and TPv of Caruana et al.’s [16] emotional laughter network are also of interest. In an earlier study, Caruana et al. were also able to elicit laughter and mirth from participants through electric stimulation of the ACC [22]. The ACC,

amygdala, and insula were also identified by another fMRI study of participants across age groups viewing humorous video stimuli [20]. This study, by Talami et al., also found the lMTG and right fusiform gyrus to show greater activation during humor detection in comparison to expression. Prenger et al. expanded research on the role of the VS and dorsal striatum (DS) in humor cognition. Through experiments that involved fMRI data from participants that listened to clips of jokes and watching a sitcom episode, they found that DS is involved in humor comprehension, while the VS is involved in both humor comprehension and appreciation [18].

Amir et al. ran another fMRI study with images that were presented with either non-humorous or humorous captions. They found activity in the temporal-parietal junction (TPJ), temporo-occipital junction (TOJ), temporal poles, and medial prefrontal cortex (mPFC) [21]. Attempting to decode crying and laughter in participants viewing naturalistic audiovisual stimuli, another study also found correlation with laughter in the STG, TPJ, IFC, and ACC. When compared with crying in this study, laughter was followed by stronger activation in the lSTG [23]. Participants in this study were not asked to laugh or cry, and it may be assumed that their reactions were involuntary and reflected emotional state. Feng et al considered the temporal fMRI activation dynamics of humor processing with regards to the detection of an incongruity, its resolution, and the subsequent mirth [24].

To my knowledge, there are no studies that explicitly attempt to disentangle the effects of background laughter on humor comprehension and appreciation. Evaluating how the auditory experience of hearing laughter while listening to something intended to be funny differs from hearing the same thing without the corresponding “laugh track” is a novel pursuit.

2.5 Audio Stories as Dynamic Naturalistic Stimuli

fMRI studies of humor have used a wide variety of stimuli and experimental designs. It is also worth discussing the choice of audio-only stories as the stimulus of choice. Naturalistic stimuli have shot up in popularity in fMRI studies in the last decade [25]. They more closely mirror what the brain goes through in real life as compared to studies with static or highly-constructed stimuli. As such, results from participants engaging with a story being told in the scanner map more readily to the experience of people listening to someone telling a story in real life.

Many older studies of humor have also exposed participants to humor through very constructed designs (for instance, asking them to rank consecutive cartoons on a funniness scale [9]). While this made sense for establishing foundational results, it is not how we experience humor in the real world. The artificial design approach leaves more open questions about the persistence of results outside the lab. Hearing jokes in within a longer story better mirrors our real life experiences of mirth as compared to reading off isolated jokes. Additionally, the fact that this data came from studies that did not ask participants to attend to the humor aspect while listening makes the responses all the more authentic.

It is also important for this study that participants only listened to the audio of the stories. They were not physically present when the story was being told, nor did they see any video of others reacting to the story. This allows for the isolation of the effect of just the sound of laughter.

Chapter 3

Data and Pre-processing

3.1 The ‘Narratives’ Dataset

As mentioned previously, I will be using a subset of the fMRI data in the ‘Narratives’ collection aggregated by Nastase et al[3]. This dataset contains 891 fMRI scans collected from over 300 subjects over several years at the Princeton Neuroscience Institute. In my analyses, I will be using the scans collected from two sets of participants who listened to two versions of the same story, ‘Pie Man’ by Jim O’Grady. The first version was recorded in front of a live audience at The Moth¹, an non-profit storytelling event. The second version was told by Mr. O’Grady at the Princeton Neuroscience Institute as he was undergoing fMRI scanning [3]. The ‘Pie Man’ story is a humorous anecdote from his time as a student reporter at Fordham University, when he wrote about a campus vigilante who smashed pie in the face of some widely disliked campus characters. It is freely available to listen to at <https://www.youtube.com/watch?v=3nZzSUDECLo>. While some bits were cut out of the PNI version, the versions shared a total 23 punchlines with each other (details on what constitutes a punchline are outlined in subsection 3.2.2).

The fMRI scans contain 3-dimensional BOLD data from a subject’s brain at some

¹<https://themoth.org/>

time point. In the ‘Narratives’ dataset, these images were taken at 1.5-second frequency (1 TR = 1.5 seconds). The live version contained 300 TRs, while the PNI version contained 294.

3.1.1 Understanding BOLD data

Individual participants brains have different anatomical dimensions. Thus raw BOLD from different participants’ brains may have varying amounts of voxels. As such, in order to enable easier generalization of analyses to multiple subjects, a few standard brain ‘templates’ are widely used in the literature. Software is available to reshape brain volumes from individual participants into these standard spaces. The details of this process are not relevant, but the ‘Narratives’ dataset provided alignments of its BOLD data to several standard spaces. I used the versions that were aligned to the widely used 2009 MNI (Montreal Neurological Institute) Nonlinear Asymmetrical template [26].

The actual BOLD data I worked with were in the NIfTI-1 (Neuroimaging Informatics Technology Initiative) [27, 28] and GIFTI² file formats. NIfTI files contain a header with various metadata and a four-dimensional array contain three axes of BOLD data per TR: (x, y, z, TR) . The value of a voxel at a time can be accessed by indexing into the its coordinates and desired TR. For the live data, dimensions were (65, 77, 49, 300) and for the studio data dimensions were (78, 93, 78, 294). The discrepancy in voxel dimensions was due to the scans being taken by two scanners with different resolutions. In order to address this issue, I up-sampled the lower resolution data to the higher dimension. I chose to do this rather than down-sampling the higher resolution data in order to preserve as much spatial precision as possible.

This was achieved using Nilearn³, a popular and powerful Python library that enables the application of machine learning methods to neuro-imaging data. In fact,

²GIFTI (RRID:SCR_009579)

³<https://nilearn.github.io/stable/index.html>

almost all the analyses in this work are done with its help. For image resampling, it offers the `resample_to_img` and `resample_img` functions. It gives a few options in terms of interpolation functions (techniques through which the image is actually transformed). I opted for default ‘continuous’ function, Nilearn’s implementation of a 3rd-order B-spline interpolation. The details of this function are not important, but it is popular in MRI interpolation due to its excellent performance [29].

GIFTI files, on the other hand, project the brain data onto a common surface ‘mesh’ of connected vertices rather than using the voxels. These surface spaces are called fsaverage templates (FreeSurfer ⁴average, named after the software package). They were adapted from the coordinate system initially published by Fischl et al [30]. The GIFTI images provided in the ‘Narratives’ dataset are aligned to the `fsaverage6` space, which contains about 40k vertices per hemisphere. Instead of being laid out in a 4-dimensional array, the information for a GIFTI volume is laid out in a simple (*vertices*, *TRs*) matrix. A single vertex at a given time point can be indexed into using its index and the TR number. However, the data is also separated by hemisphere. An advantage of the GIFTI files was that they were already aligned in the same space, with no need to down- or up-sample either one.

3.2 Pre-processing

3.2.1 Automated transcript generation and JSON formatting

While the ‘Narratives’ dataset provides extensively preprocessed fMRI data, further annotation of the stimuli themselves was required for the analyses set forth in this work. Transcripts of the audio files relevant to this study had been generated by Nas-

⁴<https://surfer.nmr.mgh.harvard.edu/>

tase et al [3] using an older software called Gentle⁵. I opted to create new transcripts using WhisperX’s large-v3 speech-to-text model[31]⁶. The transcripts themselves were stored in JSON files formatted according to Snippet 1.

This allowed for the addition of an “annotations” section to document pertinent events during the course of the story. For simplicity, I chose to add only two event types: “punchline” and “audience_laughter”. Because punchlines can come in the middle of sentences or span across multiple, this was a cleaner approach than attempted to add a field like “contain_punchline” or “contain_laughter” in a sentence segment. Two manually added fields of note for each “punchline” event are “elicits_laughter” and “shared_punchline_id”. “elicits_laughter” was marked true when a punchline was followed by laughter in the next few seconds. This field was added because not all the punchlines in the live stories were necessarily followed by audience laughter. “shared_punchline_id” was specific to the *‘Pie Man’* story and was added as a way to align shared punchlines across the live and studio versions for direct comparison.

For identifying background noise and audience laughter, I used TensorFlow’s YAMNet model for audio event class prediction⁷. This was supplemented by manual addition of shorter and quieter instances of laughter that it missed. I used a small in-browser script to obtain precise timestamps from audio files and added manual annotations to the JSON files with a python script. The onset and duration of these events likely have errors within the negligible region of tens of milliseconds.

⁵<https://github.com/strob/gentle>

⁶<https://github.com/m-bain/whisperx>

⁷<https://github.com/tensorflow/models/tree/master/research/audioset/yamnet>

```

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  "audio_filepath": "audio/pieman_audio.wav",
  "duration_sec": 412.83,
  "sample_rate_hz": 16000,
  "language": "en",

  "transcript_source": {
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    "model": "large-v3",
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  },

  "full_text": "Complete transcript text here...",

  "segments": [
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      "avg_logprob": -0.12,

      "words": [
        {
          "word": "I",
          "start": 15.127,
          "end": 15.187,
          "confidence": 0.738
        },
        ...
      ]
    }
  ],

  "annotations": {
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        "label": "punchline",
        "start": 69.218,
        "end": 71.8,
        "annotator": "manual",
        "notes": "mashing a cream pie into Dean McGowan's face."
      },
      {
        "event_id": "evt_0bf6d38e",
        "label": "audience_laughter",
        "start": 71.954,
        "end": 73.5,
        "annotator": "manual",
        "notes": ""
      },
      ...
    ]
  }
}

```

Snippet 1: Example JSON annotation file

3.2.2 What’s in a joke?

One important design question I made was determining what could be classified as a joke. Ultimately, the question I am trying to answer is whether hearing laughter in the background of something while engaging with something that is intended to elicit a humor response impacts a participant’s humor appreciation. So, guided by Chan et al.’s [14] tri-component model, I chose to define ‘punchlines’ as the inciting incidents for expected mirth. In this work, punchlines are the segments in a story when something unexpected happens, an incongruity gets resolved, an exaggeration is made, and so on. In order to find punchlines, I first attempted to prompt large-language models (LLMs) such as Gemini and Claude to identify them with context given on what should constitute a punchline. However, they ended up missing many segments that elicited audience laughter, and misclassifying segments that were clearly not intended to be humorous. So instead I opted to label all the punchline events manually. For stories with audience laughter, I first went through all the moments of audience laughter and identified the preceding segment that elicited that laughter. My criterion for the beginning of the segment was the first unexpected word. As for the end, the speaker would naturally pause in most cases after the punchline was delivered. There were also some instances of punchlines that didn’t elicit laughter, but were clearly intentional humorous writing. Intentional humorous writing was also the criterion I used for identifying punchlines in the stories without audience laughter. These identifications were also looked through by a few volunteer annotators in order to reduce bias. The idea is that across stories with and without audience laughter participants experience mirth within a second or two of hearing a punchline.

Table 3.1: Region of Interest (ROI) Definitions and Atlas Mappings

ROI	Atlas	Labels
Right Middle Frontal Gyrus	Destrieux	R G_front_middle
Left Middle Frontal Gyrus	Destrieux	L G_front_middle
Bilateral Superior Temporal Gyri	Harvard-Oxford	Superior Temporal Gyrus, anterior division; Superior Temporal Gyrus, posterior division
Left Heschl's Gyrus	Destrieux	L G_temp_sup-G_T_transv
Medial Prefrontal Cortex	Harvard-Oxford	Frontal Medial Cortex, Superior Frontal Gyrus
Left Temporal Pole	Destrieux	L Pole_temporal
Right Inferior Temporal Gyrus	Destrieux	R G_temporal_inf
Right Middle Temporal Gyrus	Destrieux	R G_temporal_middle
Ventral Striatum	Harvard-Oxford	Left Accumbens, Right Accumbens
Precuneus	Harvard-Oxford	Precuneous Cortex
Posterior Cingulate Cortex	Destrieux	L G_and_S_cingul-Mid-Post, L G_and_S_cingul-Post, R G_and_S_cingul-Mid-Post, R G_and_S_cingul-Post
Bilateral Temporoparietal Junction	Harvard-Oxford	Supramarginal Gyrus, anterior division; Supramarginal Gyrus, posterior division; Angular Gyrus
Bilateral Superior Temporal Sulcus	Destrieux	R S_temporal_sup, L S_temporal_sup

Chapter 4

Experimental Design and Implementation

4.1 Experiment I: A General Linear Model

The first experiment I set up was a general linear model (GLM), which has been used in fMRI analyses for over three decades now [32]. Albeit an older technique, it makes for a good initial analysis due to its straightforwardness. At its core, this technique attempts to fit a linear regression (estimate the linear coefficients β) for each voxel in the brain volume. It estimates the intensity of voxel at a given time point using the regressors. In this analysis, if the estimated β s predict the voxel intensity well over time, that voxel's intensity can be interpreted as having a linear relationship with the regressor over time. Details on significance will be covered in section 5.1.

4.1.1 Model Setup

First Level Model

When creating a GLM fMRI for fMRI data, the variables that need to be set up are the regressors (X), a convolution function (h), and confounds (C).

Confound variables are external variables that may be correlated with both the errors and the regressors. The confound variables common to fMRI data, such as head motion, scanner drift (a long-known phenomenon in which the signal detection gradually gets worse over the course of a scanning session [33]), low-frequency noise, and global signals that may be present throughout the entire scan have already been filtered out by Nastase et al[3]. in the cleaned versions of scans in the ‘Narratives’ dataset. As such I did not need to over-correct by re-adding these same confounds in my model.

The convolution function is necessary to account for hemodynamic lag. In fact, the convolution function in fMRI analyses is called a hemodynamic response function (HRF) [34]. As mentioned in section 2.2, hemodynamic lag is the delay in the peak of BOLD contrast. HRFs are designed to model this lag, which varies across participants. Although Lindquist et al. recommend using a change-point HRF for studies that involve an imprecise emotional response [34], their proposed method requires a lot of baseline data for comparison [35]. The event-dense dynamic stimuli used for this study are not well-suited for this method. Instead, I opted to use the default SPM (Statistical Parametric Mapping) HRF in the Nilearn library.

Last but not least, I had to determine the regressors for this model. Within each group, I set these up to be the punchline events from the JSON data. In my model, the indicator variables X_i represent whether humor comprehension and/or appreciation is occurring in the participants’ brains. The proxy I chose for this is the onsets of the punchline events in my JSON data. The hypothesis is that somewhere around the duration of the punchline, humor cognition is happening among participants. From the JSON files, I created event tables that indicated the onset and duration of each punchline event. X_i would be equal to 1 during the TRs encompassing these events and 0 otherwise. Nilearn provides a `FirstLevelModel` function that can take this as input and run a general linear model yielding the results of the regression.

The entire per-voxel formula for the first-level regression is the following:

$$y_i = \beta(X_i \cdot h) + C + \varepsilon_i$$

It is also important to address the error terms in this regression. For a GLM, there is an underlying assumption that the errors are independent and identically distributed over all samples (time points). However, this is not the case for fMRI data as the errors are highly correlated over time [36]. This makes intuitive sense: due to the repetitive nature of this biological response, a voxel’s value at time t can predict its value at time $t + 1$ well. In order to address this phenomenon, termed temporal autocorrelation, fMRI analyses use specific auto-regressive models that account for it. Nilearn provides a commonly used model called AR(1).

Snippet 2, which I applied to a randomized subset of both the live and studio groups, encompasses the GLM pipeline.

After fitting this model, a contrast map is computed comparing the estimated difference in β s when a participant hears a punchline versus not for each data point. However, it is entirely possible that these differences could’ve just come from noise in the data. In order to verify statistical *significance*, it is necessary to run a one-sample t-test. This essentially compares the calculated contrast to the distribution of possible contrasts if the true difference were 0 (the null hypothesis). The **t-statistic** itself tells us how many standard errors (an estimate of the standard deviation) our observed difference is from that null hypothesis mean. A related statistic called the **p-value** gives a percent chance that the result might be observed under the null hypothesis. A **z-score** corresponds directly with a p-value value—it just transforms the values so that they lie on a standard normal distribution, allowing for easier interpretation. This is the main statistic I will be discussing in the evaluation.

```

1 def generate_z_maps(events, subs):
2     z_maps = []
3     for sub in subs:
4         file_name = f"sub-{sub:03d}_task-{task}_space-MNI152Nlin2009cAsym_res\
5                     -native_desc-clean_bold.nii.gz"
6         file_dir = f"{cleaned_dir}sub-{sub:03d}/func/"
7         func_fn = file_dir + file_name
8         img = nib.load(func_fn)
9         print(img.shape)
10
11        glm = FirstLevelModel(
12            t_r=1.5,
13            signal_scaling=False,
14            noise_model='ar1',
15            standardize=False
16        )
17
18        glm.fit(img, events=events)
19        z_maps.append(glm.compute_contrast('punchline', output_type='z_score'))
20    return z_maps

```

Snippet 2: function for fitting a GLM and generating z-maps for NIfTI images using Nilearn. Adapted from Claude. GIFTI version in Snippet 6.

Second Level Model

After obtaining the initial results, I then ran a second-level model comparing the two groups. In this model, instead of comparing events within subjects, we are comparing conditions across sets of subjects. The differences in activation learned in the first level model become one of the inputs to the second-level model, which tries to see whether there is a significant difference in computed contrasts. Let's say there are N subjects in total, with $N/2$ in each group. The second level model can be represented by the following formula:

$$C_i = X_G \cdot [\beta_i^1, \beta_i^2] + \eta$$

Here, C_i is an N -length vector of z-scores for each participant at node i , starting with all the live condition values and followed by all the studio condition values. X_G is the

so-called design matrix, an $N \times 2$ matrix with the first $N/2$ rows of the first column set to 1 and the remaining $N/2$ rows set to 0 and vice versa in the second column. η is the error term, representing mixed effects errors and encompassing both within- and across-subject variance. The β_i scalars are the group-level betas representing the population means for the two different samples. These are the coefficients that the regression is estimating. To evaluate this model, we need to use a two-sample t-test rather than a one-sample t-test. This accounts for the fact that we’re testing the difference in means for two different β_i s. The null hypothesis is that $\beta_i^1 - \beta_i^2 = 0$ for a given node, and we can once again generate z-scores showing the likelihood of getting these β s if there was no difference between the two groups. This is all standard practice, and also easily handled in the Nilearn library using the `SecondLevelModel` function (see Snippet 5).

4.2 Experiment II: Punchline Classification Through Multi-Variate Pattern Analysis

For my second big experiment, I wanted to explore the prediction abilities of the two groups. The main idea behind this is simple: can we use the activation patterns in certain regions of interest to classify whether or not some event is a punchline? How do the two groups perform comparatively in this task? It’s important to note, however, that despite being interested in the differences in classification accuracy between groups, I did not anticipate this exploratory experiment would offer strong evidence for evaluating difference in humor cognition between the groups.

In this experiment, I leveraged the spatial structure of activity in the data (the activation pattern). This is in contrast with the first experiment, which treated each node as independent and did not take into account spatial structure whatsoever. Multi-variate pattern analysis (MVPA) encodes information about shared activity

that is lost in univariate analyses. MVPA has been a very popular technique in fMRI analysis since the early 2000s [37]. In this approach, we fit models to small clusters of voxels (ROIs) rather than estimating β s for one voxel at a time.

4.2.1 Further Preprocessing

ROI Sampling

In order to perform MVPA, we first need to isolate voxels that belong to the humor-associated ROIs from section 2.4.

The standard method for isolating brain regions is creating a brain ‘mask’ with a subset of voxels in a standard space marked as 1 and the rest marked as 0. That mask can then be applied to pre-aligned NIFTI images to select the relevant voxels in an ROI using the `NiftiMasker` function.

There are a few different way to define the ‘scope’ of an ROI. I chose to define the ones for this study using brain atlases offered by Nilearn. These atlases are just various parcellations of the brain proposed by prior research and mapped out digitally in Nilearn. No single atlas provided a parcellation that separated out all of the ROIs for this analysis, so I used masks from two standard atlases. These were the Harvard-Oxford and Destrieux [38] cortical atlases. Some details of this selection are provided in Table 3.1. I also ran all the analyses on the GIFTI data using different ROIs from the Destrieux surface atlas.

Baseline Event Sampling

In order to actually be able to perform punchline classification, I needed events to classify against. I sampled these from each subject’s BOLD time series, with the following constraints:

1. Each baseline onset is least 12 seconds from any punchline onset, avoiding HRF

overlap that would contaminate the baseline regressor with punchline signal.

2. Baseline onsets must be at least 4 seconds apart from each other, the minimum amount that would allow for an equal amount of baseline and punchline events.
3. Baseline onsets must fall at least 20 seconds from the start or end of the run.

In order to have a chance classification be at a clean 50% , I sampled the same number of baseline events as there were punchline events. Baseline events were sampled randomly for each participant.

4.2.2 Model Setup

Per-Trial Beta Estimation

Rather than one contrast map per condition, MVPA requires a vector of features per event. I estimated a separate activation pattern for every individual punchline and baseline event per participant. The standard approach would be to fit a GLM with one regressor per trial (“Least Squares All”, or LSA), but this becomes unstable when events are close together in time. The HRF-convolved regressors for adjacent events are highly collinear, which results in inflated variance for single events [39]. Since the punchline events in this dataset are close together, I instead used Mumford et al.’s **Least Squares Separate (LSS)** procedure: for each event i , I fit a GLM in which that event gets its own regressor (‘target’) and all other events are merged into a single nuisance regressor (‘others’):

$$Y = X_i^{\text{target}} \cdot \beta_i + X^{\text{others}} \cdot \beta_{\text{others}} + \varepsilon$$

Here Y is the $T \times V$ BOLD time series matrix, where T is the number of TRs and V is the number of vertices or voxels ($T \times x \times y \times z$ for the NIfTI version). As in section 4.1.1, I convolved the target regressor with the SPM HRF. The metric of

interest here is β_i : a length- V vector of estimated responses to event i across the cortical surface. Repeating this for all N punchline events and all N baseline events yields two $N \times V$ matrices of per-trial activation patterns, one for each class.

As discussed in the previous section, the data has already been cleaned, so there was no need to re-introduce other confounds as regressors in the LSS model. Temporal autocorrelation is less of a concern for LSS than for standard timeseries inference because the metrics of interest are the point estimates of β_i rather than their t-statistics. Hence I used ordinary least squares rather than the AR(1) model used in section 4.1.1.

Classification

For each ROI, I selected the corresponding voxels/vertices according to section 4.2.1. This gave me a feature matrix of shape ($\#$ trials, $\#$ vertices-in-ROI) with a binary vector of labels (punchline = 1, baseline = 0). I then trained a linear support vector machine (SVM) [40] on these features, with leave-one-out cross-validation. This process consists of a variable number of folds (in this experiments, $k=44$). In a fold, each event is held out, the classifier is trained on the remaining events, and the held-out event’s label is predicted. Accuracy is the mean of correct predictions across folds. Linear SVM is a robust and extremely popular classifier, fitting the needs of this classification. Features were z-scored within each training fold before being passed to the SVM.

For an overview, Snippet 3 shows a condensed version of this entire pipeline.

```

1  """Volumetric MVPA Pipeline Summary -- Punchline vs Baseline Classification"""
2  # --- 1. Load & mask functional data ---
3  def load_masked(nifti_path, roi_mask_3d):
4      ...
5
6  # --- 2. LSS: one GLM per trial to get single-trial beta maps ---
7  def lss_single_trial(timeseries, target_event, other_events, tr):
8      n_features, n_scans = timeseries.shape
9      events = pd.concat([
10         target_event.assign(trial_type="target"),
11         other_events.assign(trial_type="others"),
12     ], ignore_index=True)
13     dm = make_first_level_design_matrix(
14         ...
15     )
16     return betas[dm.columns.get_loc("target")] # (n_features,)
17
18 def lss_beta_series(timeseries, events_df, tr):
19     """Run LSS for every trial -> (n_trials, n_features) beta matrix."""
20     ...
21     return np.vstack(betas)
22
23 # --- 3. LOO-CV classification with linear SVM ---
24 def classify_loo(X_punch, X_base):
25     """Leave-one-out SVM accuracy for punchline (1) vs baseline (0)."""
26     X = np.vstack([X_punch, X_base])
27     y = np.array([1] * len(X_punch) + [0] * len(X_base))
28     clf = make_pipeline(StandardScaler(), SVC(kernel="linear", C=1))
29     return np.mean([
30         clf.fit(X[tr], y[tr]).predict(X[te])[0] == y[te][0]
31         for tr, te in LeaveOneOut().split(X)
32     ])
33
34 # --- 4. Per-participant pipeline ---
35 def run_participant(func_path, events_df, baseline_events, roi_masks, tr):
36     """Returns {roi_name: accuracy} for one participant."""
37     ...
38
39 # --- 5. Group-level inference: permutation test ---
40 def permutation_ttest(a, b, n_perm=10000):
41     """Non-parametric two-sample test via label shuffling."""
42     t_obs, _ = stats.ttest_ind(a, b, equal_var=False)
43     combined = np.concatenate([a, b])
44     rng = np.random.default_rng(0)
45     null_t = np.array([
46         stats.ttest_ind(
47             (p := rng.permutation(combined))[: len(a)],
48             p[len(a) :], equal_var=False,
49         )[0]
50         for _ in range(n_perm)
51     ])
52     return t_obs, np.mean(np.abs(null_t) >= np.abs(t_obs))

```

Snippet 3: Condensed volumetric MVPA pipeline. Per-trial beta patterns are estimated via Least Squares Separate (LSS), classified with a leave-one-out linear SVM, and compared across groups with a permutation test. Full implementation with parallelization, baseline sampling, confound handling, and FDR correction available at <https://github.com/Vasilam/humor-in-the-brain/>.

Chapter 5

Evaluation

5.1 Experiment I: The General Linear Model Approach

I first attempted the experiments in this section using NIfTI data, with varying success according to how I resampled the two groups to match each other. So I opted to re-run everything with the GIFTI surface files, which had the same resolution.

First Level Results

I ran the first-level analyses for randomly selected subsets of size 45 for both groups, with promising-looking differences in the activation across visualizations of individual participants in different groups. This of course has no group-level statistical significance, but these analyses do tell us whether individual participants saw effects in the punchline versus non-punchline conditions. Figure 5.1 illustrates this effect within two different participants from the different groups. With the z-score thresholded at 2.3 (corresponding to a p-value of about 0.01), we can see that the laughter condition did indeed yield an effect in both groups. We see positive correlation in the STG. With this successful first-level analyses, I moved on fitting the second-level model.

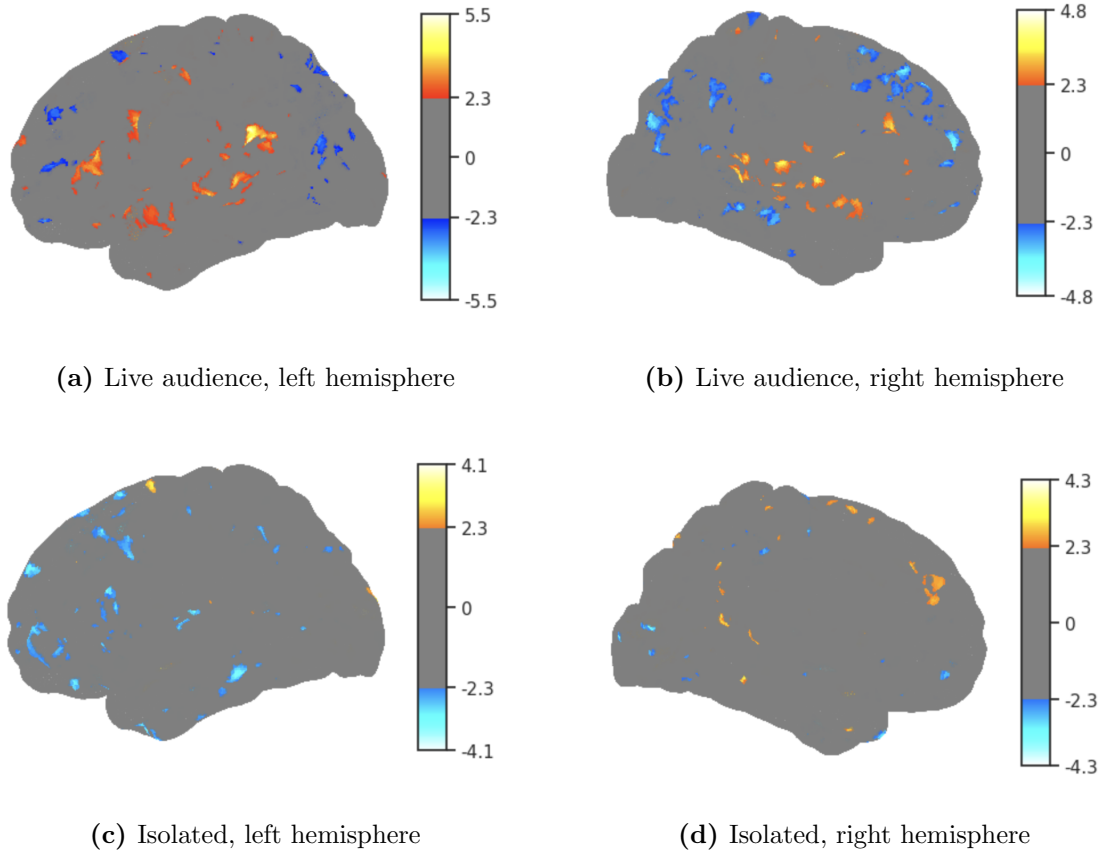


Figure 5.1: First-level GLM activation maps for punchline events. Top row: live audience group (laughter present); bottom row: isolated recording group. Warm colors indicate positive correlation, cool colors indicate negative correlation. Maps are thresholded at voxelwise uncorrected $p < .01$. Colorbars reflect each panel's t -statistic range.

Multiple Comparisons Testing

Before interpreting these results, it is important to discuss the issue of multiple comparisons. For each vertex in the brain image, we are running a two-sample t-test to check the differences in β s. This results in over 80000 simultaneous comparisons. The issue with our z-scores and p-values is that they only guarantee a threshold for each test. At a p-value 0.05, we guarantee that for each voxel there is only a 5% chance that difference could've been observed under the null hypothesis (if there were no effect). So, over the entire brain, we can expect to have around 4000 false positives! This issue is well-known in fMRI analysis, and has been illustrated in the famous dead Atlantic salmon paper. Bennett et al. demonstrated that significant differences in voxels activated in the brain of a dead Atlantic salmon, which was shown different stimuli, were found in the absence of multiple comparisons correction[41]. It's important to note that we didn't need to do this for our first-level model. At the first level, we were only estimating the effects for each voxel for individual participants. The second level is when we are actually evaluating group-level significance. Two common approaches to correction are family-wise error rate (FWER) and false discovery rate (FDR). The family-wise error rate, originally proposed by Bonferroni, involves controlling for the probability of even one false positive [42, 43]. The false discovery rate is less conservative, instead granting a bound on the proportion of false positives in significant nodes [44]. These were both used in the dead salmon paper, and are by far the most popular correction methods in GLM inference studies [41].

Keep in mind, however, that this error correction assumes a known normal distribution. An alternative way to threshold results that has gained popularity more recently is non-parametric permutation testing [45, 46]. Instead of just correcting the error rate, this involves also fitting the model hundreds or thousands of times with permuted labels. This just means we scramble the data such that random participants are labeled as having listened the live or studio version, eliminating any real signal that

might have existed in the ground truth. Running this test allows us to gauge the likelihood of getting results like ours for the whole brain from a sample size of 10000 random runs. It learns the null distribution from our own data, avoiding the assumptions in traditional parametric testing. As such, it is generally now considered to be the superior method. Learning this distribution, the Nilearn `non_parametric_inference` function returns a properly-thresholded map for the whole brain. This is not a z-map, but rather the negative logarithm of FWER-corrected p-values for each voxel. In this implementation, a threshold of 1.3 corresponds to a 5% likelihood of a single false positive across all voxels.

5.1.1 Second Level Results

As displayed in Figure 5.2, even at the most rigorous level of thresholding, we still see small clusters of statistically significant group-level differences. This is an exciting result! By parcellating the surface image of p-values into regions via the Destrieux surface atlas (Table 5.1), we see that the biggest and most significant clusters are in regions that have been associated with humor comprehension and appreciation.

There is significant activation in the lSTG, rSTG, left and right Heschl’s gyrus, bilateral transverse temporal sulcus, left precentral gyrus (primary motor cortex), and bilateral inferior circular sulcus of insula. With strong evidence for the significance of the increased activation of the STG, which has a known role in humor comprehension and appreciation, I posit that participants in the live recording group had experienced heightened humor comprehension and/or appreciation. The same goes for the left and right Heschl’s gyrus. The transverse temporal sulcus neighbors these regions, and may also be implicated in humor cognition. The significant cluster in the precentral gyrus seems to be right around the SMA, which is associated with the physical response of laughter. The roles of regions with only a couple of significant vertices are unclear, I will not attempt to interpret them.

5.1.2 Staggered Onset Analysis

You might recall from the beginning of the study that punchline onsets and durations were determined manually. In order to verify the reliability of these regressors themselves, I decided to run the FWER-corrected, non-parametric testing version of the GLM with two modified versions of the events table: one in which punchline onset labels were pushed forward by 1.5 seconds (1 TR), and another in which they were pushed back by 1.5 seconds. The resulting significance maps of the staggered onset analyses are in Figure 5.3. These results bolster confidence in the chosen punchline events. If we shift the onset 1.5 seconds before the punchline event, we see almost no significance. There are small significant clusters in lateral sulci (the folds above the STG) in both hemispheres. This could perhaps be associated with the anticipation of a punchline. Also, it's important to note that most punchline events are longer than 1.5 seconds, so both the staggered conditions still have some overlap with the punchline. In the forward-shifted version, we see less significance around the STG clusters, but a similar overall activation pattern. This makes sense—it overlaps with the actual punchline event, and we are still engaging in humor cognition around a second and a half after the joke has ended.

An interesting result from this comparison is that the pre-SMA area cluster in the precentral gyrus gets bigger and brighter (meaning a an even greater discrepancy between groups) in the forward-shifted version. When we hear a punchline, it takes us a beat to react with the physical laughter response. Though data is not available on whether or not the participants in either group were laughing at any given point, I hypothesize that this result indicates the background laughter group was laughing along to the story more than the isolated recording group.

Left hemisphere				
Region	Number of vertices	Mean p	Best p	
1 Superior temporal gyrus, lateral aspect	120	0.00249	0.0001	
2 Transverse temporal gyrus (Heschl’s gyrus)	78	0.00300	0.0002	
3 Transverse temporal sulcus	75	0.00370	0.0001	
4 Superior temporal gyrus, planum temporale	43	0.00602	0.0003	
5 Precentral gyrus	10	0.02046	0.0059	
6 Inferior circular sulcus of insula	8	0.00955	0.0020	
7 Lateral fissure, posterior ramus	2	0.03629	0.0344	
8 Lateral occipito-temporal sulcus	2	0.03248	0.0258	
Right hemisphere				
Region	Number of vertices	Mean p	Best p	
1 Superior temporal gyrus, planum temporale	44	0.00593	0.0002	
2 Transverse temporal gyrus (Heschl’s gyrus)	43	0.00141	0.0001	
3 Superior temporal gyrus, lateral aspect	43	0.00344	0.0001	
4 Transverse temporal sulcus	41	0.00047	0.0001	
5 Inferior circular sulcus of insula	2	0.00987	0.0075	

Table 5.1: Summary statistics on regions containing vertices active under family-wise error (FWER)-corrected permutation testing. Regions determined using the Destrieux surface atlas.

5.2 Experiment II: Punchline Classification Through Multi-Variate Pattern Analysis

I also attempted this experiment using both NIfTI and GIFTI data. There was no explicit computational constraint due to the mismatched volumetric resolution for the NIfTI images because metric being compared across groups was accuracy.

The leave-one-out cross-validation accuracy described in the setup was calculated per participant within each ROI separately. It reflected how well punchlines and base-lines can be distinguished in one participant’s right Heschl’s gyrus, for instance. In order to make group-level conclusions, I aggregated the accuracies across participants within groups. For each ROI, I performed a one-sample t-test with the null hypothesis being that classification accuracy was not better than chance (50%). I found that

both groups reliably classified punchline events above chance across all ROIs. The bilateral STS had the highest accuracy for the surface data classification, and the bilateral STG had the highest accuracy for volumetric data classification. This result goes well with the results of the first level from the GLM experiment, where we saw the highest correlations with these regions for both groups.

In order to compare the two groups, I ran a two-sample test at each ROI. Once again, I used permutation testing to find the right error rate to correct for. As before, this meant shuffling the group labels and eliminating any potential signal. I opted for FDR correction, but found that even with the weakest bounds for significance the results did not differ between the live and studio groups. However, I unexpectedly found that using surface data yielded much better accuracy than using volumetric data! I am not entirely sure why this was the case, because for both volumes there are more actual data points in the volumetric data (about 250k voxels) as opposed to the surface data (around 80k vertices)

5.3 Limitations

There are several limitations that should be addressed in this work.

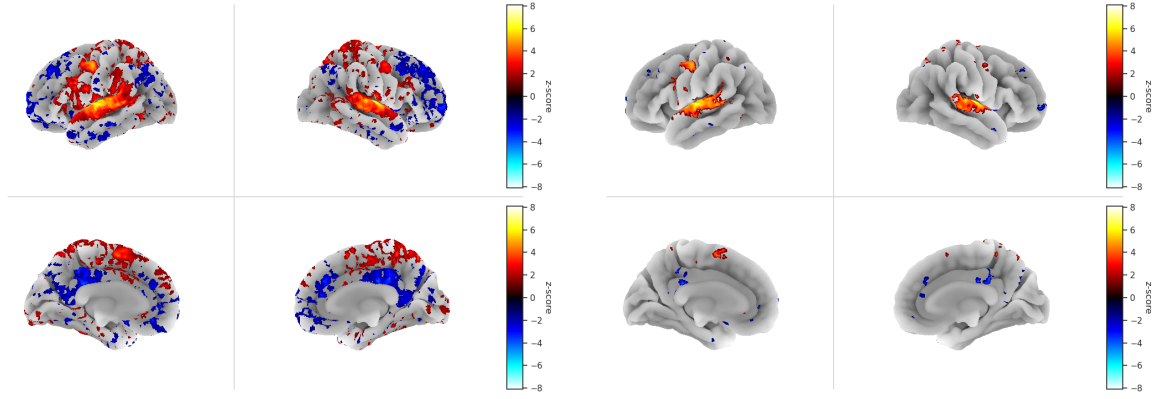
The first is the fact that the data for the two groups comes from two different sets of subjects. In an ideal scenario for consistency, the same participants would be scanned twice — once listening to the story with background laughter and once without. Instead, two independent groups of subjects were scanned on different years and using different scanners. Even though we got significant results despite this (likely a result of the large sample size), it is possible some smaller yet true differences got over-corrected out. While the fMRI data was aligned to standard templates and the lower-resolution volumetric data was up-sampled, residual differences in scan quality and resolution may have introduced noise into the analyses. The discrepancy in

voxel dimensions between the two groups ($65 \times 77 \times 49$ for the live condition versus $78 \times 93 \times 78$ for the isolated condition) is a direct consequence of this.

The two versions of the story are also not perfectly identical. While both recordings share 23 punchlines, the isolated version had some segments cut out, making the data nonviable for TR-by-TR analyses, such as a shared response model-based one. Differences in pace and delivery between a live performance and an in-scanner recording may also have independently influenced neural responses, confounding the effect that is attributed to background laughter alone.

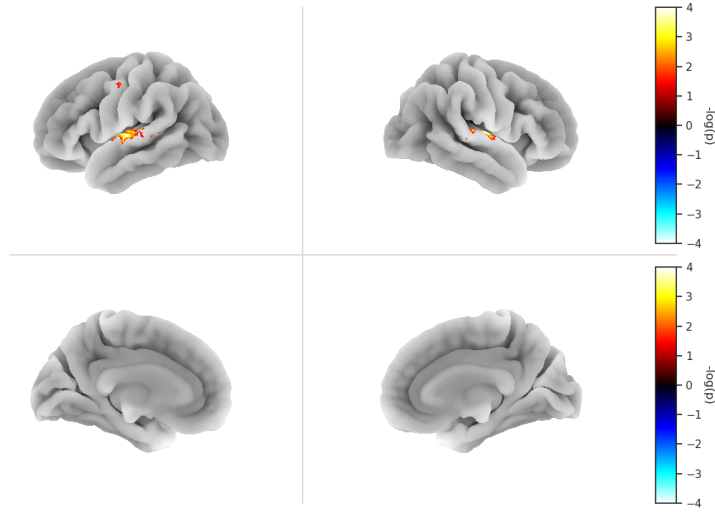
Punchline annotation was performed manually, which introduces subjectivity. Although annotations were reviewed by volunteers to reduce individual bias, the boundaries of each punchline event — particularly their onset and offset — are still approximate. The volunteer base was also comprised mainly of college students, who may share similar senses of humor. Onset and offset timestamps may contain errors on the order of tens of milliseconds, and other annotators may not have agreed on the precise definition of what constitutes a punchline. The staggered onset analysis in subsection 5.1.2 provides some reassurance about the robustness of the chosen events, but does not eliminate this concern entirely.

No behavioral or physiological data were collected during or after scanning to confirm whether participants were actually laughing or experiencing mirth. The inference that punchline events corresponded to humor cognition rests on the assumption that participants responded to the story like a general audience. In particular, the hypothesized difference in laughter behavior between the two groups (subsection 5.1.2) cannot be confirmed without the actual data on that. Finally, the study relies on a single story for the direct between-group comparison. While ‘Pie Man’ is a well-received humor-forward story, findings based on one story and one speaker may not generalize to other types of humor, other storytelling styles, or other forms of background laughter such as canned laugh tracks versus genuine audience responses.



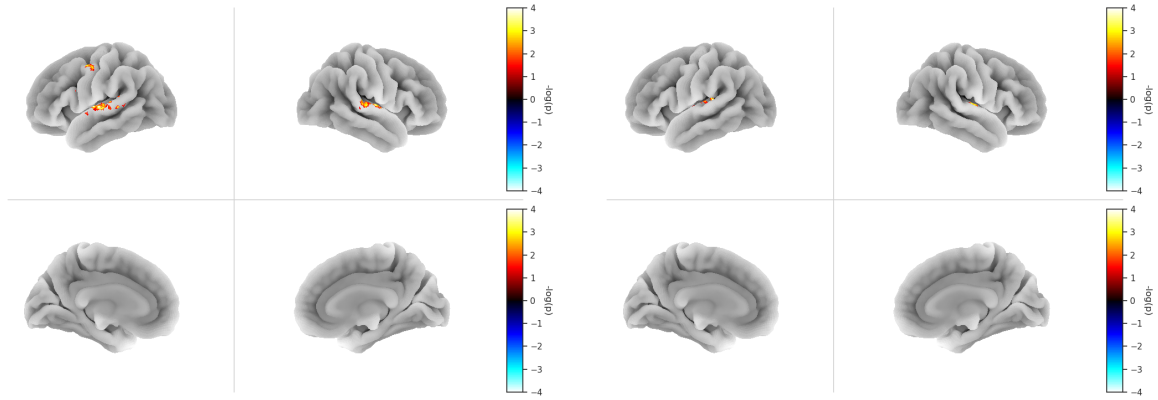
(a) Uncorrected.

(b) Parametric FDR-corrected.



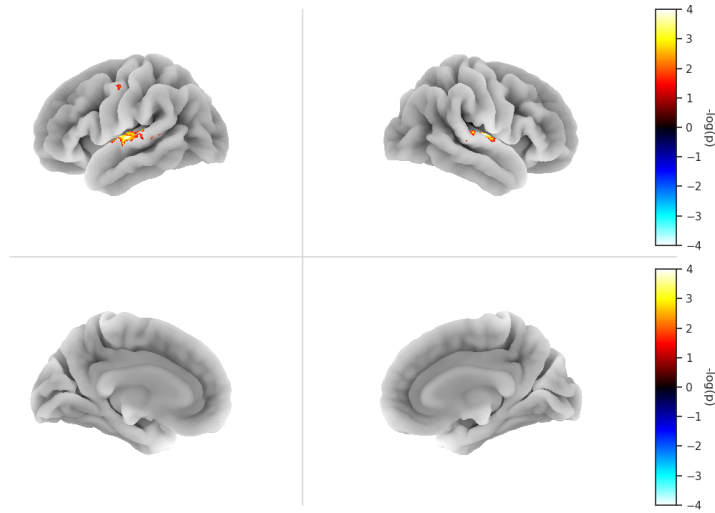
(c) Permutation-based non-parametric FWER correction.

Figure 5.2: Statistical surface maps under different multiple-comparison corrections. In each panel, the top row shows lateral views of the left and right hemispheres, and the bottom row shows medial views. (a) is uncorrected, only thresholded at $p < 0.05$. (b) is parametrically FDR-corrected with rate 0.05. (c) is non-parametrically FWER-corrected based on permutation testing with a 5% likelihood of a single false positive. Even under the most stringent error-correction, we see statistically significant differences in the left and right superior temporal gyri, both associated with humor comprehension.



(a) 1.5s after punchline onset.

(b) 1.5s before punchline onset.



(c) At punchline onset.

Figure 5.3: Statistical surface maps (permutation testing FWER-corrected $-\log(p)$ values, 10000 permutations) at staggered punchline onset times (1.5 second intervals). In each panel, the top row shows lateral views of the left and right hemispheres, and the bottom row shows medial views. (a) is one TR after each punchline onset. (b) is one TR before each punchline onset. (c) is at the punchline onset. Most punchlines last longer than a TR, so there is some overlap in the events used in (a) and (b) with the baseline onset (c).

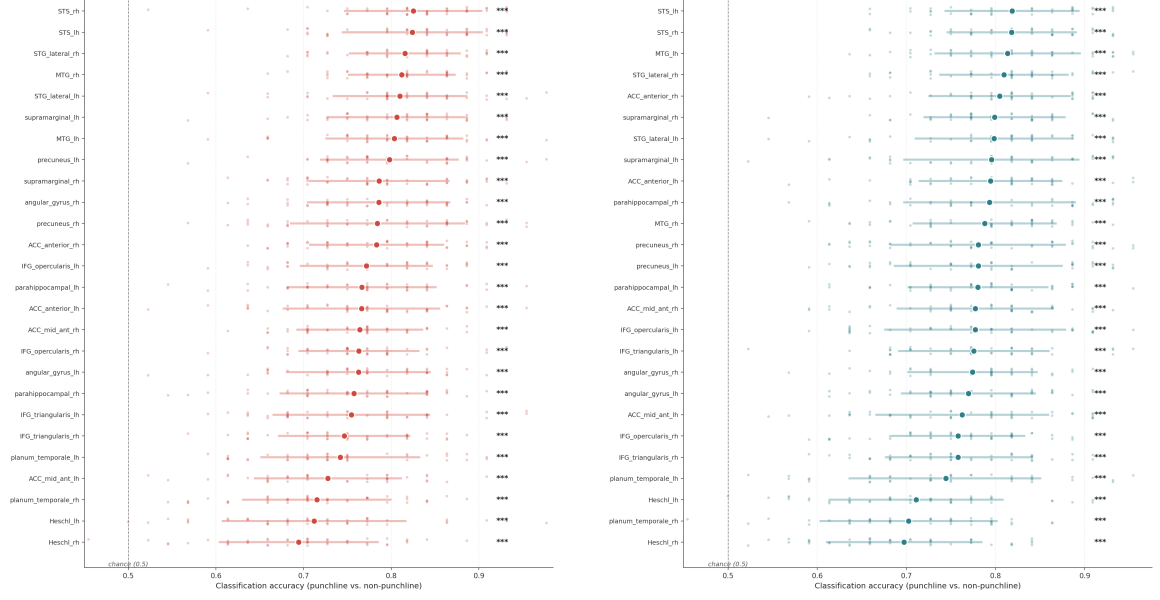


Figure 5.4: Surface-based (GIFTI) MVPA classification accuracy for punchline vs. non-punchline events across regions of interest. Large dots indicate group means, horizontal bars denote ± 1 SD, and small dots show individual participants ($N = 45$ per group). The dashed vertical line marks chance performance (0.5). Stars indicate significance of one-sample t -tests against chance ($*p < .05$, $**p < .01$, $***p < .001$). All ROIs decoded significantly above chance in both groups.

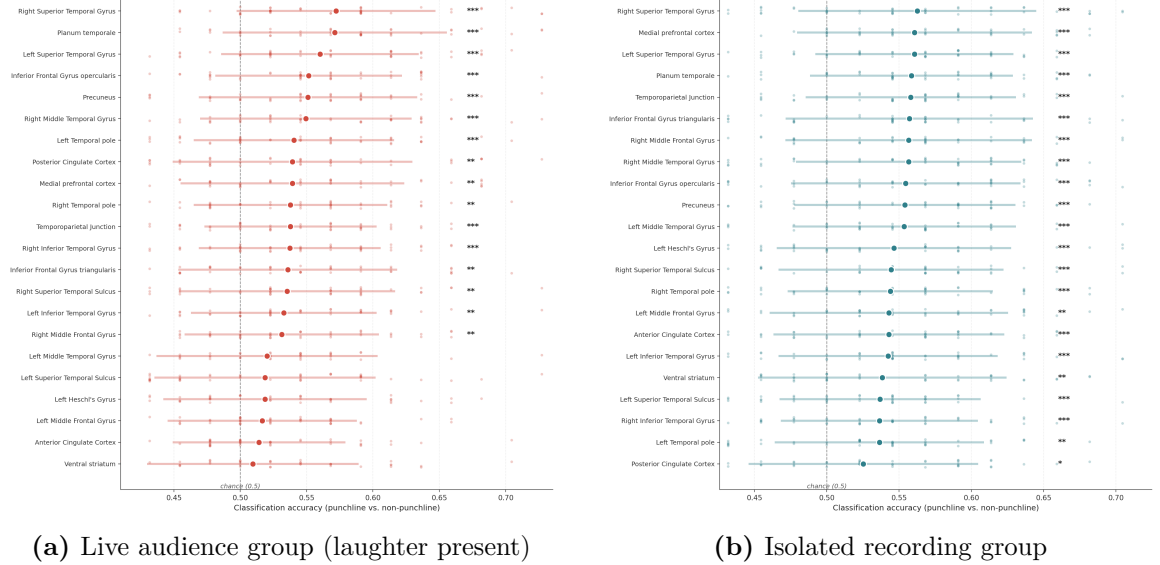


Figure 5.5: Volumetric (NIFTI) MVPA classification accuracy for punchline vs. non-punchline events across regions of interest. Conventions as in Figure 5.4. Decoding accuracies were substantially lower than in the surface-based analysis, though most temporal and frontal ROIs still classified significantly above chance.

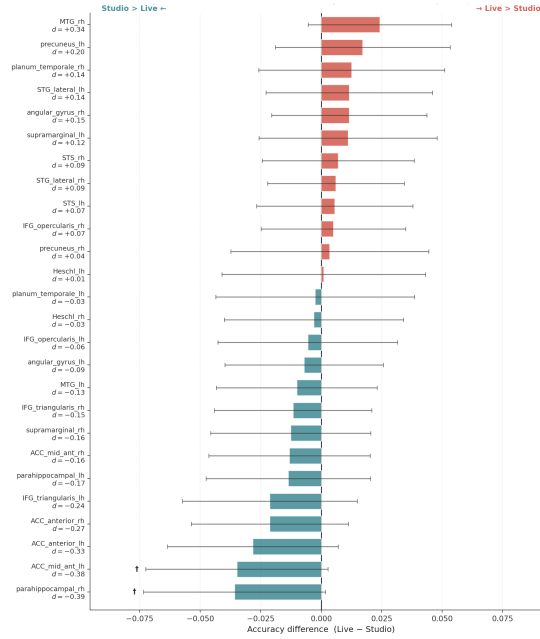


Figure 5.6: Between-group differences in surface-based MVPA classification accuracy (Live – Studio). Positive values (coral) indicate higher decoding accuracy in the live audience group; negative values (teal) indicate higher accuracy in the studio group. Error bars represent approximate 95% confidence intervals. Cohen's d is reported beneath each ROI label. Significance markers reflect uncorrected and FDR-corrected between-group t -tests († $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$). No ROI survived FDR correction; two ROIs showed uncorrected trend-level differences ($p < .10$) favoring the studio group.

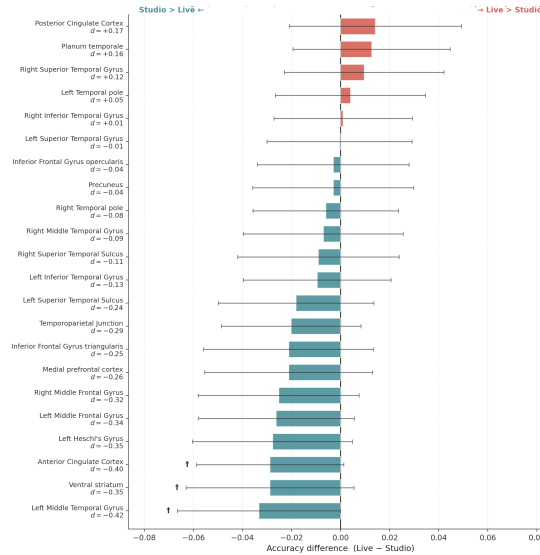


Figure 5.7: Between-group differences in volumetric MVPA classification accuracy (Live – Studio). Conventions as in Figure 5.6. No ROI reached FDR-corrected significance; three ROIs (anterior cingulate cortex, ventral striatum, and left middle temporal gyrus) showed uncorrected trend-level differences ($p < .10$) favoring the studio group.

Chapter 6

Conclusions and Future Work

In this work, I leveraged fMRI data analysis techniques to explore differences in brain activation patterns between two sets of participants that listened to versions of the same story. One set heard audience laughter in the background, while the other heard only the narrator’s voice. The goal of this analysis was to find out whether hearing audience laughter in the background of the media we are consuming impacts our humor comprehension and appreciation in the brain. I manually classified punchline events in both versions of the story, and ran analyses on punchlines shared between the two.

The first experiment I ran was a general linear model comparison between the two groups. Through this, I tested whether the background laughter group experienced a statistically significant increase in activation. Under gold standard family-wise error rate (FWER) correction under nonparametric permutation testing, which ensured only a 5% chance of a single false discovery, I found significantly higher activation in clusters around the superior temporal gyri, Heschl’s gyri, and the pre-semimotor area (Figure 5.2). These results supported my hypothesis. The left and right STGs and Heschl’s gyri have all been determined as neural correlates of humor comprehension and appreciation in prior studies. The pre-SMA has been identified multiple times as

a region involved in the physical response of laughter.

In order to test the reliability of my punchline events, I repeated this analysis with the punchline onsets staggered 1.5 seconds forwards and backwards. For the backwards-staggered version, there was less of significant difference between conditions, and it was constrained to a much smaller area. Almost all punchline events were longer than 1.5 seconds, so the small significant difference might be explained by the overlap with the actual punchline event. For the forwards-staggered version, I observed less difference in the STG-area clusters, and greater difference in pre-SMA area cluster. There is no data on whether participants were laughing during their scans, but I hypothesize this heightened delayed pre-SMA response might point to the background laughter group laughing along to the story after the punchline more than their counterparts. Overall, these results (Figure 5.3) corroborate the reliability of the chosen punchline events.

My second experiment went beyond analyzing independent voxel activation, using multi-variate pattern analysis to perform punchline classification. This type of analysis considers regions of interest and their joint patterned activity. For each subject, I randomly selected approximately punchline-length baseline events from the story. I then trained classifiers to differentiate between the baseline and punchline events using the estimated patterns at each ROI. I evaluated their performance using leave-one-out cross-validation and found that both groups were able to classify punchline events with above-chance accuracy across ROIs with no significant difference in classification accuracy between the two groups. Recall that the data was available in two formats: a 3-dimensional volumetric representation (NIFTI), and a 1-dimensional set of connected vertices (GIFTI). I ran the MVPA-based classification with both sets of data, and found GIFTI was much better at the classification task.

One potential future direction for this work is to address the data limitations faced in this study. The ideal experimental setup would be to have the same set of

subjects scanned in the same machine listen to two versions of a humorous story, one with background laughter and one without. The stories should follow the same exact script (which was not the case for our Pie Men) and be almost perfectly aligned by TRs. Positive results in such an analysis would more reliably support the conclusions made in this work.

From a software perspective, it would also be very useful to generate new brain surface maps that are able to parcellate higher-resolution brain surface images (the currently available Destrieux atlas only support around 20k vertices, meanwhile modern brain surface images have up to 164k vertices).

Other future directions could involve applying other novel fMRI analysis techniques to further explore the data. One such possibility is applying functional connectivity analyses to explore how brain regions work in tandem during punchline events, and whether that connectivity is strengthened under the laughter condition.

Yet another interesting question is the impact of background laughter when something is not funny. This study has shown that hearing background laughter in a well-written and acclaimed short story with humorous beats results in heightened activity in brain areas associated with mirth, but would hearing a laugh track in the background of something unfunny actually elicit a negative response? What about when the listener can tell the laughter isn't authentic? These questions would better illustrate how social auditory cues shape our emotional experience, and suggest that the brain's response to laughter is not just reflexive but also context-sensitive.

Appendix A: Code

All the code for this study can be found in GitHub: <https://github.com/VasilaM/humor-in-the-brain> . Some relevant snippets are attached here.

Adaptation Disclosure: much of the initial setup for file access was adapted from NEU480: fMRI Decoding: Reading Minds Using Brain Scans, taught Spring 2026 by Prof. Kenneth Norman and Zaid Zada.

AI Usage Disclosure: a large proportion of the code was initially generated by LLMs (Claude, ChatGPT, Gemini). All AI-generated code was thoroughly inspected, tested, and modified when necessary. AI also assisted the research process through answering field-specific questions, finding relevant studies (all vetted), helping generate Latex figures, and other similar tasks.

```

1 def generate_z_maps_gifti(events, subs, task):
2     """
3     Run first-level GLM on GIFTI surface data.
4     Both hemispheres are combined into a single SurfaceImage per subject.
5
6     Returns a list of SurfaceImage z-maps, one per subject.
7     """
8     z_maps = []
9
10    for sub in subs:
11        gifti_imgs = {}
12        for hemi, side in [('L', 'left'), ('R', 'right')]:
13            file_name = (
14                f"sub-{sub:03d}_task-{task}_space-fsaverage6"
15                f"_hemi-{hemi}_desc-clean.func.gii"
16            )
17            file_dir = f"{cleaned_dir}sub-{sub:03d}/func/"
18            gifti_imgs[side] = file_dir + file_name # pass path, not loaded image
19
20            surf_img = SurfaceImage(
21                mesh=fsaverage6_mesh,
22                data=gifti_imgs # {'left': path_L, 'right': path_R}
23            )
24
25            print(f"sub-{sub:03d}: {surf_img.shape}")
26
27            glm = FirstLevelModel(
28                t_r=1.5,
29                signal_scaling=False,
30                noise_model='ar1',
31                standardize=False
32            )
33
34            glm.fit(surf_img, events=events)
35            z_maps.append(glm.compute_contrast('punchline', output_type='z_score'))
36
37    return z_maps

```

Snippet 4: function for fitting a first-level GLM and generating z-maps for GIFTI images using Nilearn. Adapted from Claude.

```

1 def run_second_level_gifti(z_maps_group1, z_maps_group2,
2                             label_group1='live', label_group2='pni',
3                             alpha=0.05):
4     """
5     Run a second-level two-sample t-test on surface z-maps.
6
7     Parameters:
8     z_maps_group1 : list of SurfaceImage
9     First-level z-maps for group 1 (one per subject)
10    z_maps_group2 : list of SurfaceImage
11    First-level z-maps for group 2 (one per subject)
12
13    Returns a dict with z-maps for: group1 > group2, group2 > group1,
14    and each group's one-sample effect
15    """
16
17    # Combine z-maps and build design matrix
18    all_z_maps = z_maps_group1 + z_maps_group2
19    n1, n2 = len(z_maps_group1), len(z_maps_group2)
20
21    design_matrix = pd.DataFrame({
22        label_group1: [1] * n1 + [0] * n2,
23        label_group2: [0] * n1 + [1] * n2,
24    })
25
26    # Fit second level model
27    second_level_model = SecondLevelModel(n_jobs=-1)
28    second_level_model.fit(all_z_maps, design_matrix=design_matrix)
29
30    # Contrasts
31    contrasts = {
32        f'{label_group1}_gt_{label_group2}': f'{label_group1} - {label_group2}',
33        f'{label_group2}_gt_{label_group1}': f'{label_group2} - {label_group1}'
34    }
35
36    results = {}
37    results_thresholded = {}
38    for name, contrast_def in contrasts.items():
39        results[name] = second_level_model.compute_contrast(
40            contrast_def,
41            output_type='z_score'
42        )
43        # Apply FDR correction
44        thresholded_map, threshold = threshold_stats_img(
45            results[name],
46            alpha=alpha,
47            height_control='fdr'
48        )
49        results_thresholded[name] = thresholded_map
50
51    return results, results_thresholded

```

Snippet 5: function for fitting a second-level GLM and generating z-maps for GIFTI images using Nilearn. Adapted from Claude.

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